

Dong-u Shin

Mechanical Engineering Graduate | sea7yy@naver.com

EDUCATION

Sunmoon University — B.S., Mechanical Engineering

Mar 2020 — Feb 2026

Total GPA 4.1/4.5 · Major GPA 4.3/4.5 · Asan-si, Chungcheongnam-do, South Korea

JOURNAL

A Proof of Concept Framework for an Integrated Decision-Making Subprocessor Linking Image-Based Modeling, Simulation Databases, and LLM Reasoning

2026

Dong-u Shin, Jungho Seo, Yonghui Park | Journal of Mechanical Science and Technology · Accepted, 2026

CONFERENCE

Development of a Digital Twin Platform for Automated Structural Analysis Using Image Analysis and Large Language Models for Predicting Plant System Status

2026

Jungho Seo, Jun Yeok Park, Dong-u Shin, Yonghui Park | 2026 KSME Plant Engineering Division Spring Conference, Jeju, Korea · May 27-28, 2026

기계 시스템 동적 응답 해석의 효율성 향상을 위한 웹 기반 모델링 및 해석 플랫폼 개발

2025

Jungho Seo, Dong-u Shin, Yonghui Park | 2025 KSEE Annual Conference, Daegu, Korea · September 22-23, 2025

RESEARCH EXPERIENCE

Computational Mechanics Lab (CML), Sunmoon University — Undergraduate Researcher

Mar 2025 — Nov 2025

Integrated image-based modeling and simulation-command databases with LLM-based engineering decision support. Developed a web-based modeling and simulation workflow for dynamic response analysis of mechanical systems.

AWARDS

Excellence in Presentation Award (Poster Session)

Sep 2025

2025 KSEE Annual Conference · 기계 시스템 동적 응답 해석의 효율성 향상을 위한 웹 기반 모델링 및 해석 플랫폼 개발

CERTIFICATION

Engineer General Machinery

Sep 2025